

Linear Models Lecture 4: Algebra of Bias

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Why Partition $\mathbf{X}$?

- In applied work, we rarely care equally about every regressor. We typically have:
 - A **treatment** or variable of interest ($\mathbf{X}_1$), and
 - **Controls** we include to avoid omitted variable bias ($\mathbf{X}_2$).
- Partitioning $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$ lets us answer three questions:
 - 1 What is the formula for $\hat{\beta}_1$ *holding $\mathbf{X}_2$ constant*? → Frisch-Waugh-Lovell.
 - 2 What happens to $\hat{\beta}_1$ if we *omit $\mathbf{X}_2$* ? → Omitted variable bias formula.
 - 3 How sensitive is $\hat{\beta}_1$ to *unobserved* confounders? → Cinelli-Hazlett sensitivity analysis.
- The algebra of partitioned matrices (Schur complement) is the common tool behind all three results.

Partitioned Regression

- We have seen that it is possible to partition the matrix $\mathbf{X}$ and β into subcomponents:

$$\mathbf{y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \mathbf{e}$$

$$\mathbf{X} = [\mathbf{X}_1 \quad \mathbf{X}_2]$$

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} \mathbf{X}_1' \\ \mathbf{X}_2' \end{bmatrix} [\mathbf{X}_1 \quad \mathbf{X}_2] = \begin{bmatrix} \mathbf{X}_1'\mathbf{X}_1 & \mathbf{X}_1'\mathbf{X}_2 \\ \mathbf{X}_2'\mathbf{X}_1 & \mathbf{X}_2'\mathbf{X}_2 \end{bmatrix}$$

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

- We used this partition to study how including (or failing to include) control variables affects a multivariate regression.

Why Open Up $(X'X)^{-1}$?

- One of the virtues of OLS is that it never hides where its answers come from. Every coefficient can be traced back to specific variation in the data.
- When you run a multivariate regression, the estimate of β_1 does *not* use all the variation in X_1 . It uses only the part of X_1 that the other regressors cannot explain.
- The next few slides prove this with matrix algebra. The derivation is the hardest in the course—you do not need to memorize the steps.
- What you need to take away: the Schur complement tells you exactly which variation in the data identifies each coefficient. That transparency is why we study this estimator.

Formula for Inverse of $\mathbf{X}'\mathbf{X}$

Here is what we are going to derive:

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} (\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1} & -(\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1} \\ -(\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}_1\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1(\mathbf{X}'_1\mathbf{X}_1)^{-1} & (\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}_1\mathbf{X}_2)^{-1} \end{bmatrix}$$

Where

$$\mathbf{P}_1 = \mathbf{X}_1(\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1$$

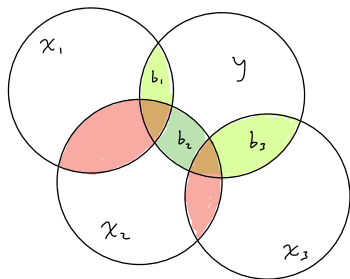
$$\mathbf{P}_2 = \mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2$$

Schur Complement: A 2×2 Example

- Before diving into the general derivation, let's see what the formula does with two scalar regressors (education, experience): $\mathbf{X}'\mathbf{X} = \begin{pmatrix} 10 & 3 \\ 3 & 8 \end{pmatrix}$
- Direct inversion: $(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{10 \cdot 8 - 3^2} \begin{pmatrix} 8 & -3 \\ -3 & 10 \end{pmatrix} = \frac{1}{71} \begin{pmatrix} 8 & -3 \\ -3 & 10 \end{pmatrix}$
- Schur complement for B_{11} : $\left(10 - \frac{3^2}{8}\right)^{-1} = \left(\frac{71}{8}\right)^{-1} = \frac{8}{71} \checkmark$
- The denominator $10 - \frac{9}{8} = \frac{71}{8}$ is the variance of education *not explained by* experience.

The Schur complement isolates the *unique* information in $\mathbf{X}_1$ after removing $\mathbf{X}_2$. More collinearity $\rightarrow$ less unique info $\rightarrow$ larger standard errors.

Intuition: What Does the Schur Complement Remove?



$$\begin{aligned}
 (\mathbf{X}_1' \mathbf{X}_1 - \mathbf{X}_1' \mathbf{P}_2 \mathbf{X}_1)^{-1} &= (\mathbf{X}_1' \mathbf{X}_1 - \mathbf{X}_1' \mathbf{P}_2 \mathbf{P}_2 \mathbf{X}_1)^{-1} \\
 &= \left(\underbrace{\mathbf{X}_1' \mathbf{X}_1}_{\text{variance of } \mathbf{X}_1} - \underbrace{\mathbf{X}_1' \mathbf{P}_2 \mathbf{P}_2 \mathbf{X}_1}_{\text{Projection of } \mathbf{X}_1 \text{ onto the column space of } \mathbf{X}_2} \right)^{-1}
 \end{aligned}$$

Finding Inverse of Partitioned Matrix

Call the $k \times k$ partitioned matrix $\mathbf{A}$ and its inverse $\mathbf{A}^{-1}$, $\mathbf{B}$:

$$\begin{bmatrix} \mathbf{X}'_1 \mathbf{X}_1 & \mathbf{X}'_1 \mathbf{X}_2 \\ \mathbf{X}'_2 \mathbf{X}_1 & \mathbf{X}'_2 \mathbf{X}_2 \end{bmatrix} = \mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix}$$

We know that if $\mathbf{B} = \mathbf{A}^{-1}$, then $\mathbf{AB} = \mathbf{I}$

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_k & \mathbf{0}_{k,n-k} \\ \mathbf{0}_{n-k,k} & \mathbf{I}_{n-k} \end{pmatrix} \end{aligned}$$

Finding $\mathbf{B}_{11}$: Setting Up Two Equations

The top-left and bottom-left blocks of $\mathbf{AB} = \mathbf{I}$ give us two equations in two unknowns ($\mathbf{B}_{11}$ and $\mathbf{B}_{21}$):

$$\mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} = \mathbf{I}_k \quad \text{(from top-left block)}$$

$$\mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} = \mathbf{0}_{n-k,k} \quad \text{(from bottom-left block)}$$

Strategy: solve the second equation for $\mathbf{B}_{21}$, then substitute into the first.

Finding B_{11} : Substitution

$$B_{21} = -A_{22}^{-1} A_{21} B_{11} \quad (\text{solve 2nd equation for } B_{21})$$

$$A_{11} B_{11} + A_{12} [-A_{22}^{-1} A_{21} B_{11}] = I_k \quad (\text{substitute into 1st equation})$$

$$(A_{11} - A_{12} A_{22}^{-1} A_{21}) B_{11} = I_k \quad (\text{factor out } B_{11})$$

The expression $A_{11} - A_{12} A_{22}^{-1} A_{21}$ is the **Schur complement**—it will turn out to be the residual variance of X_1 after removing X_2 .

Inverse of Partitioned Matrix (finding B_{11})

$$(\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})\mathbf{B}_{11} = \mathbf{I}_k$$

$$(\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1} = \mathbf{B}_{11}$$

$$(\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1)^{-1} = \mathbf{B}_{11}$$

$$(\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1} = \mathbf{B}_{11}$$

(inverse of Schur complement)

(plugging in for $\mathbf{A}$ s)

(Sub in notation for $\mathbf{P}_2$.)

Result: Formula for Inverse of Partitioned Matrix

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix}$$

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} (\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1} & -(\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1} \\ -(\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}_1\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1(\mathbf{X}'_1\mathbf{X}_1)^{-1} & (\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}_1\mathbf{X}_2)^{-1} \end{bmatrix}$$

Using Inverse of Partitioned Matrix formula

Regressing $\mathbf{X}_1$ onto $\mathbf{X}$ gives us the identity matrix and 0:

$$\begin{aligned}
 (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}_1 &= (\mathbf{X}'\mathbf{X})^{-1} \begin{bmatrix} \mathbf{X}'_1\mathbf{X}_1 \\ \mathbf{X}'_2\mathbf{X}_1 \end{bmatrix} \\
 &= \begin{bmatrix} \mathbf{B}_{11}\mathbf{X}'_1\mathbf{X}_1 + \mathbf{B}_{12}\mathbf{X}'_2\mathbf{X}_1 \\ \mathbf{B}_{21}\mathbf{X}'_1\mathbf{X}_1 + \mathbf{B}_{22}\mathbf{X}'_2\mathbf{X}_1 \end{bmatrix} \\
 &= \begin{bmatrix} (\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_1 - (\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1 \\ -(\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}'_1\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1(\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_1 + (\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}'_1\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1 \end{bmatrix} \\
 &= \begin{bmatrix} (\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1)^{-1}[\mathbf{X}'_1\mathbf{X}_1 - \mathbf{X}'_1\mathbf{P}_2\mathbf{X}_1] \\ -(\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}'_1\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1 + (\mathbf{X}'_2\mathbf{X}_2 - \mathbf{X}'_2\mathbf{P}'_1\mathbf{X}_2)^{-1}\mathbf{X}'_2\mathbf{X}_1 \end{bmatrix} \\
 &= \begin{bmatrix} \mathbf{I} \\ \mathbf{0} \end{bmatrix}
 \end{aligned}$$

Frisch Waugh Lovell in Matrix Terms

- FWL claim is that the regression coefficient $\hat{\beta}_2$ is the same as the result of first regressing $\mathbf{y}$ and $\mathbf{X}_2$ on $\mathbf{X}_1$ and then on one another. That is, we can first project $\mathbf{y}$ into the orthogonal complement of the column space of $\mathbf{X}_1$, then project it onto $\mathbf{X}_2$.

$$\mathbf{y} = \mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2\hat{\beta}_2 + e \quad (\text{full model})$$

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{X}_2\hat{\beta}_2 + \mathbf{M}_1e \quad (\text{premultiply by } \mathbf{M}_1; \text{ note } \mathbf{M}_1\mathbf{X}_1 = 0)$$

$$\hat{\beta}_2 = ((\mathbf{M}_1\mathbf{X}_2)' \mathbf{M}_1\mathbf{X}_2)^{-1} (\mathbf{M}_1\mathbf{X}_2)' \mathbf{M}_1\mathbf{y} \quad (\text{OLS on residualized data})$$

$$= (\mathbf{X}_2' \mathbf{M}_1 \mathbf{X}_2)^{-1} \mathbf{X}_2' \mathbf{M}_1 \mathbf{y} \quad (\mathbf{M}_1 \text{ symmetric and idempotent})$$

- Where $\mathbf{M}_1\mathbf{q}$ produces the residuals of any variable $\mathbf{q}$ regressed on $\mathbf{X}_1$

FWL Proof Part I: Deriving $\hat{\beta}_1$

Start from the normal equations $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$ and take the first row:

$$(\mathbf{X}'_1\mathbf{X}_1)\hat{\beta}_1 + (\mathbf{X}'_1\mathbf{X}_2)\hat{\beta}_2 = \mathbf{X}'_1\mathbf{y} \quad \text{(1st normal equation)}$$

$$(\mathbf{X}'_1\mathbf{X}_1)\hat{\beta}_1 = \mathbf{X}'_1\mathbf{y} - (\mathbf{X}'_1\mathbf{X}_2)\hat{\beta}_2 \quad \text{(isolate } \hat{\beta}_1 \text{)}$$

$$\hat{\beta}_1 = \underbrace{(\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{y}}_{\hat{\beta}_{res}} - \underbrace{(\mathbf{X}'_1\mathbf{X}_1)^{-1}(\mathbf{X}'_1\mathbf{X}_2)}_{\text{OVB correction}}\hat{\beta}_2 \quad \text{(premultiply by } (\mathbf{X}'_1\mathbf{X}_1)^{-1} \text{)}$$

$$\hat{\beta}_1 = (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1(\mathbf{y} - \mathbf{X}_2\hat{\beta}_2) \quad \text{(factor: } \hat{\beta}_1 \text{ from residualized } \mathbf{y} \text{)}$$

FWL Proof Part II: Deriving $\hat{\beta}_2$ (Setup)

Start from the second row of the normal equations and substitute $\hat{\beta}_1$:

$$(\mathbf{X}_2' \mathbf{X}_1) \hat{\beta}_1 + (\mathbf{X}_2' \mathbf{X}_2) \hat{\beta}_2 = \mathbf{X}_2' \mathbf{y} \quad \text{(2nd normal equation)}$$

$$(\mathbf{X}_2' \mathbf{X}_1) [(\mathbf{X}_1' \mathbf{X}_1)^{-1} \mathbf{X}_1' (\mathbf{y} - \mathbf{X}_2 \hat{\beta}_2)] + (\mathbf{X}_2' \mathbf{X}_2) \hat{\beta}_2 = \mathbf{X}_2' \mathbf{y} \quad \text{(plug in } \hat{\beta}_1 \text{ from Part I)}$$

$$\mathbf{X}_2' \mathbf{P}_1 \mathbf{y} - \mathbf{X}_2' \mathbf{P}_1 \mathbf{X}_2 \hat{\beta}_2 + (\mathbf{X}_2' \mathbf{X}_2) \hat{\beta}_2 = \mathbf{X}_2' \mathbf{y} \quad \text{(recognize } \mathbf{P}_1 \text{)}$$

FWL Proof Part II: Deriving $\hat{\beta}_2$ (Result)

Continuing—collect $\hat{\beta}_2$ terms on the left, everything else on the right:

$$(\mathbf{X}_2' \mathbf{X}_2 - \mathbf{X}_2' \mathbf{P}_1 \mathbf{X}_2) \hat{\beta}_2 = \mathbf{X}_2' \mathbf{y} - \mathbf{X}_2' \mathbf{P}_1 \mathbf{y} \quad (\text{rearrange})$$

$$\mathbf{X}_2' (\mathbf{I} - \mathbf{P}_1) \mathbf{X}_2 \hat{\beta}_2 = \mathbf{X}_2' (\mathbf{I} - \mathbf{P}_1) \mathbf{y} \quad (\text{factor out } (\mathbf{I} - \mathbf{P}_1))$$

$$\mathbf{X}_2' \mathbf{M}_1 \mathbf{X}_2 \hat{\beta}_2 = \mathbf{X}_2' \mathbf{M}_1 \mathbf{y} \quad (\mathbf{I} - \mathbf{P}_1 = \mathbf{M}_1)$$

$$\hat{\beta}_2 = (\mathbf{X}_2' \mathbf{M}_1 \mathbf{X}_2)^{-1} \mathbf{X}_2' \mathbf{M}_1 \mathbf{y} \quad (\text{solve for } \hat{\beta}_2)$$

This is the FWL result: $\hat{\beta}_2$ from the full regression equals the coefficient from regressing $\mathbf{M}_1 \mathbf{y}$ on $\mathbf{M}_1 \mathbf{X}_2$ —residual on residual.

Frisch-Waugh-Lovell (FWL)

- Under $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$,

$$\hat{\mathbf{y}} = b_0 + b_1\mathbf{x}_1 + \dots + b_{k-1}\mathbf{x}_{k-1} + b_k\mathbf{x}_k + b_{k+1}\mathbf{x}_{k+1} + \dots + b_n\mathbf{x}_n$$

$$\tilde{\mathbf{y}} = d_0 + d_1\mathbf{x}_1 + \dots + d_{k-1}\mathbf{x}_{k-1} + 0 + d_{k+1}\mathbf{x}_{k+1} + \dots + d_n\mathbf{x}_n$$

$$\hat{\mathbf{x}}_k = c_0 + c_1\mathbf{x}_1 + \dots + c_{k-1}\mathbf{x}_{k-1} + 0 + c_{k+1}\mathbf{x}_{k+1} + \dots + c_n\mathbf{x}_n$$

- FWL: The regression coefficient b_k is equivalent to a regression coefficient b_1^* produced by regressing the residualised outcome $\mathbf{e}_y = \mathbf{y} - \tilde{\mathbf{y}}$ on the residualised $\mathbf{x}_k$: $\mathbf{e}_{x_k} = \mathbf{x}_k - \hat{\mathbf{x}}_k$.

$$\mathbf{e}_y = b_0^* + b_1^*\mathbf{e}_{x_k} + \mathbf{e}$$

FWL in R: Residual-on-Residual Regression

```
mod0 <- lm(prestige ~ education + income + women, data=Prestige)
mod1a <- lm(prestige ~ income + women, data = Prestige)
mod1b <- lm(education ~ income + women, data = Prestige)
eprest <- lm(resid(mod1a)~resid(mod1b))
coef(eprest)
(Intercept) resid(mod1b)
2.445994e-15 4.362425
coef(mod0)
(Intercept)      education      income      women
-7.524222154    4.362424649    0.001172269   -0.012946077
```

Applications Frisch-Waugh-Lovell (FWL)

- Practical: Plotting data/ coefficients from multivariate regression in 2d.
- Theoretical: Basis for sensitivity tests to evaluate the effects of omitted variables.
- Pedagogical: Improving understanding of the linear model.

Trace of Projection and Annihilator Matrix

- Recall, the **trace** of a matrix is the sum of the diagonal elements and the sum of the eigenvalues.
- The **rank** of a matrix is the maximum number of linearly independent column vectors.
- The trace of $\mathbf{P}$ is its rank k
 - The projection matrix is **idempotent**: $\mathbf{P} = \mathbf{P}^2$.
 - If λ is an eigenvalue of $\mathbf{P}$, $\mathbf{P}\mathbf{v} = \lambda\mathbf{v}$
 - Applying $\mathbf{P}$ again: $\mathbf{P}^2\mathbf{v} = \mathbf{P}(\mathbf{P}\mathbf{v}) = \mathbf{P}(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$.
 - But $\mathbf{P}^2 = \mathbf{P}$, so $\lambda^2\mathbf{v} = \mathbf{P}\mathbf{v} = \lambda\mathbf{v}$, giving $\lambda^2 = \lambda$. Only true for $\lambda = 0$ or $\lambda = 1$.
 - The eigenvalue 1 corresponds to a direction preserved by the projection, the dimension of the column space of $\mathbf{X}$.
- $tr(\mathbf{M}) = tr(\mathbf{I}) - tr(\mathbf{P}) = n - k$

Alternative proof that Trace of Projection is k

- Theorem: $\text{tr}\mathbf{P} = k$.

$$\begin{aligned}\text{tr}\mathbf{P} &= \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \\ &= \text{tr}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) \\ &= \text{tr}(\mathbf{I}_k) \\ &= k\end{aligned}$$

Leverage

- The i 'th diagonal element of $\mathbf{P}$ is $h_{ii} = X_i'(\mathbf{X}'\mathbf{X})^{-1}X_i$ is called the *leverage* of the i 'th observation.
- h_{ii} ranges between 0 and 1, measures how unusual the i^{th} observation X_i is relative to other observations.
- A regression design is called **balanced** when the leverage values are roughly equal.
- Recall, the sum of $h_{ii} = k$
- A regression is perfectly balanced if $\max(h_{ii}) = k/n$

Leave-One-Out Regression

- The Leave-One-Out Regression estimates the model excluding observation i :

$$\hat{\beta}_{(-i)} = \left(\sum_{j \neq i} \mathbf{x}_j \mathbf{x}_j' \right)^{-1} \left(\sum_{j \neq i} \mathbf{x}_j y_j \right) = (\mathbf{X}'\mathbf{X} - \mathbf{x}_i \mathbf{x}_i')^{-1} (\mathbf{X}'\mathbf{y} - \mathbf{x}_i y_i)$$

- $\hat{\beta}_{(-i)}$ is not a function of (Y_i, X_i) , so it can be used for prediction: $\tilde{Y}_i = \mathbf{x}_i' \hat{\beta}_{(-i)}$.
- The **leave-one-out residual** (prediction error) is $\tilde{e}_i = Y_i - \tilde{Y}_i$.

Leave-One-Out Regression

- Calculating $\hat{\beta}_{(-i)}$ comes for free from our projection model:

$$\hat{\beta}_{(-i)} = \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i\tilde{e}_i$$

$$\tilde{e}_i = (1 - h_{ii})^{-1}\hat{e}_i$$

- We can define a modified annihilator matrix:

$$\mathbf{M}^* \equiv (\mathbf{I}_n - \text{diag}\{h_{11}, \dots, h_{nn}\})^{-1}$$

Which allows us to rewrite:

$$\tilde{\mathbf{e}} = \mathbf{M}^*\hat{\mathbf{e}}$$

- We observe the residuals $\hat{\mathbf{e}}$ and we know how much each observation affects the regression, so we can just subtract that out.

Deriving the Leave-One-Out Formula (Step 1)

Goal: express $\hat{\beta}_{(-i)}$ in terms of $\hat{\beta}$ and $\tilde{e}_i$. Start from the definition:

$$\hat{\beta}_{(-i)} = (\mathbf{X}'\mathbf{X} - X_iX_i')^{-1}(\mathbf{X}'\mathbf{y} - X_iY_i) \quad (\text{definition})$$

$$(\mathbf{X}'\mathbf{X} - X_iX_i')\hat{\beta}_{(-i)} = \mathbf{X}'\mathbf{y} - X_iY_i \quad (\text{multiply both sides by } (\cdots))$$

$$\mathbf{X}'\mathbf{X}\hat{\beta}_{(-i)} - X_iX_i'\hat{\beta}_{(-i)} = \mathbf{X}'\mathbf{y} - X_iY_i \quad (\text{distribute})$$

$$\hat{\beta}_{(-i)} - (\mathbf{X}'\mathbf{X})^{-1}X_iX_i'\hat{\beta}_{(-i)} = \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}X_iY_i \quad (\text{premultiply by } (\mathbf{X}'\mathbf{X})^{-1})$$

Deriving the Leave-One-Out Formula (Step 2)

Continuing from the previous slide:

$$\begin{aligned}\hat{\beta}_{(-i)} &= \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i Y_i + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i \mathbf{X}_i' \hat{\beta}_{(-i)} && \text{(move } \hat{\beta}_{(-i)} \text{ terms right)} \\ &= \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i (Y_i - \mathbf{X}_i' \hat{\beta}_{(-i)}) && \text{(factor out } (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i) \\ &= \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i \tilde{e}_i && \text{(recognize } \tilde{e}_i = Y_i - \mathbf{X}_i' \hat{\beta}_{(-i)})\end{aligned}$$

The leave-one-out estimate equals the full-sample estimate minus a correction proportional to the prediction error $\tilde{e}_i$.

Connecting $\hat{e}_i$ to the Prediction Error $\tilde{e}_i$ (Setup)

Start from the LOO result. Premultiply by X_i' to get predictions:

$$\hat{\beta}_{(-i)} = \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}X_i'\tilde{e}_i \quad (\text{from previous slide})$$

$$X_i'\hat{\beta}_{(-i)} = X_i'\hat{\beta} - X_i'(\mathbf{X}'\mathbf{X})^{-1}X_i'\tilde{e}_i \quad (\text{premultiply by } X_i')$$

$$X_i'\hat{\beta}_{(-i)} = X_i'\hat{\beta} - h_{ii}\tilde{e}_i \quad (\text{recognize } h_{ii} = X_i'(\mathbf{X}'\mathbf{X})^{-1}X_i')$$

Connecting $\hat{e}_i$ to the Prediction Error $\tilde{e}_i$ (Result)

Now subtract both sides from Y_i :

$$Y_i - X_i' \hat{\beta}_{(-i)} = Y_i - X_i' \hat{\beta} + h_{ii} \tilde{e}_i \quad (\text{subtract from } Y_i)$$

$$\tilde{e}_i = \hat{e}_i + h_{ii} \tilde{e}_i \quad (\text{recognize } \tilde{e}_i \text{ and } \hat{e}_i)$$

$$\tilde{e}_i (1 - h_{ii}) = \hat{e}_i \quad (\text{collect } \tilde{e}_i \text{ terms})$$

$$\tilde{e}_i = (1 - h_{ii})^{-1} \hat{e}_i \quad (\text{solve for } \tilde{e}_i)$$

High-leverage observations (h_{ii} close to 1) inflate the prediction error relative to the residual.

Check your understanding

- For what observation would $\hat{\beta}_{(-i)} = \hat{\beta}$?

Influential observations

- An observation i is influential if its omission changes the parameter of interest.
- $\hat{\beta} - \hat{\beta}_{(-i)} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i\tilde{e}_i$.
- Premultiply by $\mathbf{X}_i'$, and we get that

$$\hat{Y}_i - \tilde{Y}_i = \mathbf{X}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i\tilde{e}_i = h_{ii}\tilde{e}_i$$

- So i is influential if h_{ii} is big and $|\tilde{e}_i|$ is big.
- This warrants investigation: it could indicate a data entry error, an outlier from a different population, or a genuine extreme case that deserves scrutiny.

Leave-One-Out in Practice

```
library(carData)
mod <- lm(prestige ~ education +
          income + women, data=Prestige)
# Leverage values
h <- hatvalues(mod)
which.max(h) # highest-leverage obs
# Studentized residuals
rst <- rstudent(mod)
# Flag: |rstudent| > 2
Prestige$flag <- abs(rst) > 2
```

- `hatvalues()` returns the diagonal of P .
- `rstudent()` gives $\tilde{e}_i / \hat{\sigma}_{(-i)}$: the leave-one-out residual scaled by its SE.
- Both high leverage *and* $|rstudent| > 2$ warrants investigation.

High leverage + large residual = investigate. It may be a data error, an outlier, or the most interesting observation in your dataset.

What Happens When We Leave Something Out?

- So far: we derived $\hat{\beta}$ assuming we included the right regressors.
- In practice, we always leave something out. The question is whether it matters.
- An estimator is **biased** if $E[\hat{\beta}|\mathbf{X}] \neq \beta$ —on average across samples, we get the wrong answer.
- The FWL result gives us the tool to quantify this: we already showed that $\hat{\beta}_1$ depends on what else is in the model. Now we ask: what if $\mathbf{X}_2$ *should* be in the model but isn't?

Example: Factory Closures and Voting Behavior

- A factory closure is visible evidence that trade may harm the local economy. Does it push voters toward protectionism?
- Suppose we regress voting on factory closures:

$$Y_i = \alpha + \beta \text{Closure}_i + \epsilon_i$$

- But closures don't happen at random—they concentrate in regions with specific *industrial compositions*. And industrial composition independently affects voting.
- The full model should include both:

$$Y_i = \alpha + \beta \text{Closure}_i + \gamma \text{IndComp}_i + \delta (\text{Closure}_i \times \text{IndComp}_i) + \epsilon_i$$

- If we omit IndComp, our estimate of β absorbs part of γ . How much?

The Omitted Variable Bias Formula

Recall the normal equations with $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$:

$$\begin{bmatrix} \mathbf{X}'_1 \mathbf{X}_1 & \mathbf{X}'_1 \mathbf{X}_2 \\ \mathbf{X}'_2 \mathbf{X}_1 & \mathbf{X}'_2 \mathbf{X}_2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{X}'_1 \mathbf{y} \\ \mathbf{X}'_2 \mathbf{y} \end{bmatrix}$$

Derivation of $\hat{\beta}_1$:

$$(\mathbf{X}'_1 \mathbf{X}_1) \hat{\beta}_1 + (\mathbf{X}'_1 \mathbf{X}_2) \hat{\beta}_2 = \mathbf{X}'_1 \mathbf{y}$$

$$(\mathbf{X}'_1 \mathbf{X}_1) \hat{\beta}_1 = \mathbf{X}'_1 \mathbf{y} - (\mathbf{X}'_1 \mathbf{X}_2) \hat{\beta}_2$$

$$\hat{\beta}_1 = (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{y} - (\mathbf{X}'_1 \mathbf{X}_1)^{-1} (\mathbf{X}'_1 \mathbf{X}_2) \hat{\beta}_2$$

$$\hat{\beta}_{res} = \hat{\beta}_1 + \underbrace{(\mathbf{X}'_1 \mathbf{X}_1)^{-1} (\mathbf{X}'_1 \mathbf{X}_2) \hat{\beta}_2}_{\widehat{\text{bias for excluding } X_2}}$$

Unpacking the Bias: Imbalance $\times$ Impact

$$\widehat{\text{bias}} = \underbrace{(\mathbf{X}_1' \mathbf{X}_1)^{-1} (\mathbf{X}_1' \mathbf{X}_2)}_{\text{Imbalance: how well does } X_1 \text{ predict } X_2} \times \underbrace{\hat{\beta}_2}_{\text{Impact: effect of } X_2 \text{ on } Y}$$

- $(\mathbf{X}_1' \mathbf{X}_1)^{-1} (\mathbf{X}_1' \mathbf{X}_2)$ are the coefficients from regressing $\mathbf{X}_2$ on $\mathbf{X}_1$ —how much do the confounders differ across treatment groups?
- $\hat{\beta}_2$ is how much X_2 matters for Y , holding X_1 constant.
- Bias is large only when *both* are large: a confounder must be imbalanced *and* consequential.
- The problem: $\mathbf{X}_1$ and $\mathbf{X}_2$ are multivariate, so the bias is a matrix product that is hard to sign. We need a scalar summary.

From Matrix Bias to a Scalar: Partial R^2

- Cinelli and Hazlett (2020) propose studying omitted variables using R^2 as a metric.
- Suppose the full linear regression is as follows:

$$Y = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \mathbf{e}$$

- $\mathbf{P}_1 = \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'$ is the projection matrix onto $\mathbf{X}_1$.
- $\mathbf{M}_1 = \mathbf{I} - \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'$ is the residual maker for $\mathbf{X}_1$.
- In general the partial R^2 measures the proportion of the variance in Y that is uniquely explained by a set of predictors $\mathbf{X}_2$ after accounting for $\mathbf{X}_1$.

Formula for R^2 with one variable

$$R^2_{Y \sim \mathbf{Z}} = \frac{SSR}{TSS} \quad (\text{definition (1) of } R^2)$$

$$= \frac{\text{var}(\hat{Y})}{\text{var}(Y)} \quad (SSR \text{ is variance explained})$$

$$= 1 - \frac{\text{var}(e)}{\text{var}(Y)} \quad (\text{since } \text{var}(Y) = \text{var}(\hat{Y}) + \text{var}(e))$$

$$= \text{corr}(Y, \hat{Y})^2 \quad (\text{definition (2) of } R^2)$$

$$= \text{corr}(Y, \mathbf{Z})^2 \quad (\hat{Y} \text{ is linear prediction from } \mathbf{Z})$$

Example in R

```
library(carData)
attach(Prestige)
mod <- lm(prestige ~ education)
summary(mod)$r.square
# [1] 0.7228007
var(predict(mod)) / var(prestige)
1 - var(resid(mod)) / var(prestige)
cor(prestige, predict(mod))^2
cor(prestige, education)^2
```

Formula for partial R^2

■ Using Residual Sum of Squares:

- $R^2_{Y \sim X_2 | X_1} = 1 - \frac{RSS(X_1, X_2)}{RSS(X_1)}$
- $RSS(X_1) = \|M_1 Y\|^2 = Y M_1' M_1 Y = Y' M_1 Y$
- $RSS(X_1, X_2) = \|M_{1,2} Y\|^2 = Y' M_{1,2} Y$

■ Using Sum of Squared Residuals:

- $R^2_{Y \sim X_2 | X_1} = \frac{SSR(X_2 | X_1)}{RSS(X_1)}$
- SSR, the explained sum of squares, the amount explained by X_2 once accounting for X_1

$$\begin{aligned} SSR(X_2 | X_1) &= Y' M_1 X_2 (X_2' M_1 X_2)^{-1} X_2' M_1 Y \\ &= Y' M_1 X_2 \hat{\beta}_2 \end{aligned}$$

Cinelli Hazlett (2020) formulation

We model $Y = \hat{\beta}_1 D + e$, so $\hat{\beta}_1 = \frac{\text{cov}(D, Y)}{\text{var}(D)}$.

What if there is regression of Y on D and there is a single omitted variable Z ?

$$\begin{aligned} \widehat{\text{bias}} &= (\mathbf{X}_1' \mathbf{X}_1)^{-1} (\mathbf{X}_1' \mathbf{X}_2) \hat{\beta}_2 \\ &= \left(\frac{\text{cov}(D, Z)}{\text{var}(D)} \right) \left(\frac{\text{cov}(Z^{\perp D}, Y^{\perp D})}{\text{var}(Z^{\perp D})} \right) \end{aligned}$$

Lets write this in terms of D and two new formalisms for the disturbance: $Z^{\perp D}$ the part of Z not predicted by D , and $Y^{\perp D}$ the part of Y not predicted by D .

Bias in Terms of Covariances

We start with the regression model:

$$Y = \hat{\beta}_1 D + e, \quad \hat{\beta}_1 = \frac{\text{cov}(D, Y)}{\text{var}(D)}.$$

Now consider the case where a variable Z is omitted from the regression of Y on D . The omitted variable bias is given by:

$$\widehat{\text{bias}} = \frac{\text{cov}(D, Z)}{\text{var}(D)} \cdot \frac{\text{cov}(Z^{\perp D}, Y^{\perp D})}{\text{var}(Z^{\perp D})}.$$

Note: The first fraction reflects how D and Z co-move, while the second captures the effect of Z on Y once D 's influence is removed.

Converting Covariances to Correlations

For the First Term:

$$\frac{\text{cov}(D, Z)}{\text{var}(D)} = \frac{\text{corr}(D, Z) \text{sd}(D) \text{sd}(Z)}{\text{sd}(D)^2} = \frac{\text{corr}(D, Z) \text{sd}(Z)}{\text{sd}(D)}.$$

Note: We used the identity

$$\text{cov}(X, Y) = \text{corr}(X, Y) \text{sd}(X) \text{sd}(Y),$$

and noted that $\text{var}(D) = \text{sd}(D)^2$.

For the Second Term:

$$\frac{\text{cov}(Z^{\perp D}, Y^{\perp D})}{\text{var}(Z^{\perp D})} = \frac{\text{corr}(Z^{\perp D}, Y^{\perp D}) \text{sd}(Z^{\perp D}) \text{sd}(Y^{\perp D})}{\text{sd}(Z^{\perp D})^2} = \frac{\text{corr}(Z^{\perp D}, Y^{\perp D}) \text{sd}(Y^{\perp D})}{\text{sd}(Z^{\perp D})}.$$

Note: Similarly, we express the covariance in terms of correlation and standard deviations, with $\text{var}(Z^{\perp D}) = \text{sd}(Z^{\perp D})^2$.

The Bias Formula in Terms of Partial R^2

Combining the two unpacked terms, we obtain:

$$\widehat{\text{bias}} = \left(\frac{\text{corr}(D, Z) \text{sd}(Z)}{\text{sd}(D)} \right) \left(\frac{\text{corr}(Z^{\perp D}, Y^{\perp D}) \text{sd}(Y^{\perp D})}{\text{sd}(Z^{\perp D})} \right).$$

Rearranging, this becomes:

$$\widehat{\text{bias}} = \frac{\text{corr}(D, Z) \text{corr}(Z^{\perp D}, Y^{\perp D}) \text{sd}(Z) \text{sd}(Y^{\perp D})}{\text{sd}(D) \text{sd}(Z^{\perp D})}.$$

Replacing partial R^2 :

$$|\widehat{\text{bias}}| = \sqrt{\frac{R_{D \sim Z}^2 R_{Y \sim Z|D}^2}{1 - R_{D \sim Z}^2}} \cdot \frac{\text{sd}(Y^{\perp D})}{\text{sd}(D)}.$$

What Does the Bias Formula Tell Applied Researchers?

- The bias depends on two quantities, both expressed as partial R^2 :
 - 1 $R^2_{D \sim Z}$: How strongly does the omitted variable predict the **treatment**?
 - 2 $R^2_{Y \sim Z|D}$: How strongly does the omitted variable predict the **outcome**, after accounting for treatment?
- If *either* partial R^2 is small, the bias is small—a confounder must predict both D and Y to be dangerous.
- In practice, you ask: “Is it plausible that an unobserved variable explains $x\%$ of the residual variation in D and $y\%$ of the residual variation in Y ?”
- Observed covariates serve as benchmarks: if the strongest observed predictor has $R^2_{Y \sim Z|D} = 0.05$, an omitted confounder would need to be far stronger to overturn your result.

Example: Simulation Setup for Sensitivity Analysis

Simulation Design:

- Generate N observations with treatment D , observed confounder Z , and outcome:

$$Y = \beta_D D + \beta_Z Z + e, \quad e \sim N(0, 1).$$

- We first show that omitting Z biases the estimate of β_D .
- Then we include Z and ask: how robust is our estimate to *additional* unobserved confounders?

Step 1: Simulate Data and Compare Models

```
set.seed(123)
N <- 1000; beta_D <- 2; beta_Z <- 3
D <- rbinom(N, 1, 0.5); Z <- rnorm(N)
Y <- beta_D * D + beta_Z * Z + rnorm(N)

# Naive model (omitting Z) -- biased
coef(lm(Y ~ D))          # beta_D != 2

# Full model (including Z) -- unbiased
model_full <- lm(Y ~ D + Z)
coef(model_full)         # beta_D ~ 2
```

The naive model is biased because Z is correlated with both D and Y . The full model recovers the true effect—but what if there were *another* confounder we could not observe?

Step 2: Running the Sensitivity Analysis

```
library(sensemkr)

# Sensitivity analysis on the FULL model
sensitivity <- sensemakr(model_full, treatment = "D",
                        benchmark_covariates = "Z")

summary(sensitivity)
plot(sensitivity)
```

Key question: How strong would an *unobserved* confounder need to be—relative to the observed confounder Z —to explain away the estimated effect of D ?

Interpreting the Sensitivity Analysis Output

Key Diagnostics:

- **Robustness Value:** The minimum strength (in terms of partial R^2) that an unobserved confounder must have with both the treatment and outcome to fully explain away the estimated effect.
- **Benchmark Comparison:** Using the observed Z as a benchmark helps gauge whether such a confounder is plausible. If an omitted variable would need to be $3\times$ stronger than Z , the result is likely robust.
- **Contour Plot:** Visualizes all combinations of $R^2_{D \sim U|X}$ and $R^2_{Y \sim U|D,X}$ that would reduce the estimated effect to zero.